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Statistical Analysis of Indian Premier League Match Data

A Case Study in Descriptive Statistics, Probability Modelling and Regression

Case Study submitted to Dr. Keerthy A. S.

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Contents

1	Introduction	1
1.1	Dataset and Variables	1
2	Descriptive Statistics	1
2.1	Central Tendency	1
2.2	Dispersion	2
3	Probability Modelling	3
3.1	The Binomial Model for First-Innings Advantage	3
3.2	A Probability Mass Function for Wicket Margins	4
3.3	Range-Based Probabilities	5
4	Regression Analysis	5
5	Conclusion and Reflection	5

1 Introduction

The Indian Premier League (IPL) is one of the world's most heavily followed cricket tournaments, and every edition produces a rich record of match outcomes. The dataset analysed here is derived from the supplied IPL match archive and records, for every league match, the two competing teams, the toss, the match venue, the target set by the team batting first, the manner and margin of victory, and the winner.

1.1 Dataset and Variables

The full archive spans 1,169 matches from 2008 to 2025. For this study we restrict attention to the ten-year window **2008–2018** (seasons 1–11), yielding 696 matches. After removing 10 abandoned or no-result matches the working sample contains **686** decided matches with a single tied match retained, giving **10** selected variables drawn from the original 19:

Table 1: The ten variables retained for analysis

Variable	Type	Role
team1, team2	Categorical	Competing teams
season	Discrete	IPL season code (2008 = 1)
toss_winner	Categorical	Team winning the toss
toss_decision	Binary	bat or field
winner	Categorical	Match winner
venue	Categorical	Stadium
result	Categorical	runs / wickets / tie
result_margin	Numeric	Margin of victory (runs or wickets)
target_runs	Numeric	Runs set by the first-batting side

The objectives are (i) to summarise the population of matches using measures of central tendency and dispersion; (ii) to model the win process with the *Binomial* distribution, the focus of the course curriculum; (iii) to model the wicket-margin variable through its probability mass function and to evaluate range-based probabilities; and (iv) to test whether first-innings targets have risen over the decade using simple linear regression.

2 Descriptive Statistics

This section summarises the two chief numeric variables: the target set by the first-batting side ($n = 686$) and the margin of victory in decided matches ($n = 685$; the single tie has no margin).

2.1 Central Tendency

For *target runs* the mean is 160.54, the median 163, and the mode 166 runs. The mean and median are close, indicating that first-innings targets are broadly symmetric about a typical value near the 160–165 run mark. For *victory margins* the mean is 17.08 and the median only 8; a mean well above the median signals a distribution dragged upward by a modest number of very large margins.

2.2 Dispersion

The standard deviation of targets is 32.41 runs and that of margins 21.95 runs. Because the two variables differ in scale, the coefficient of variation is more informative: margins have a coefficient of variation 1.285 — more than six times that of targets (0.202). In practical terms, targets are tightly clustered around 160, whereas match margins are highly volatile.

The quartiles confirm the asymmetry. For targets the interquartile range spans 142–182 (IQR 40); for margins the IQR spans 6–19 (IQR 13), with a maximum of 146. The skewness of margins is 2.81, strongly positive, reflecting abundant narrow wins and rare blow-outs. Targets are only mildly negatively skewed (-0.46).

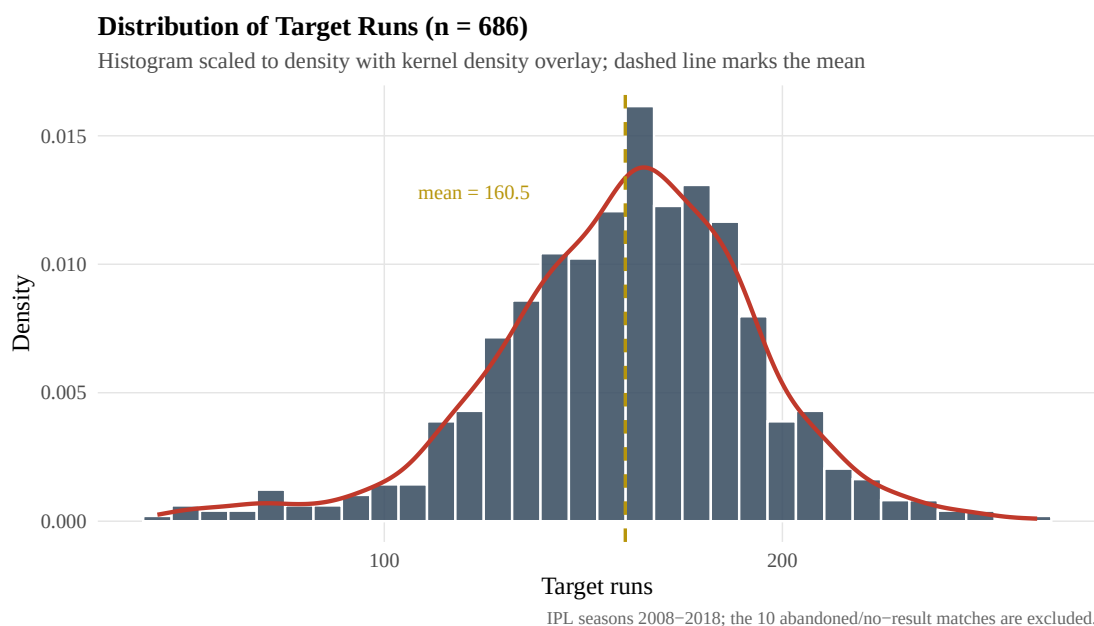


Figure 1: Distribution of first-innings target runs (n = 686). The histogram is scaled to a density and overlaid with a kernel density estimate; the dashed line marks the mean of 160.54.

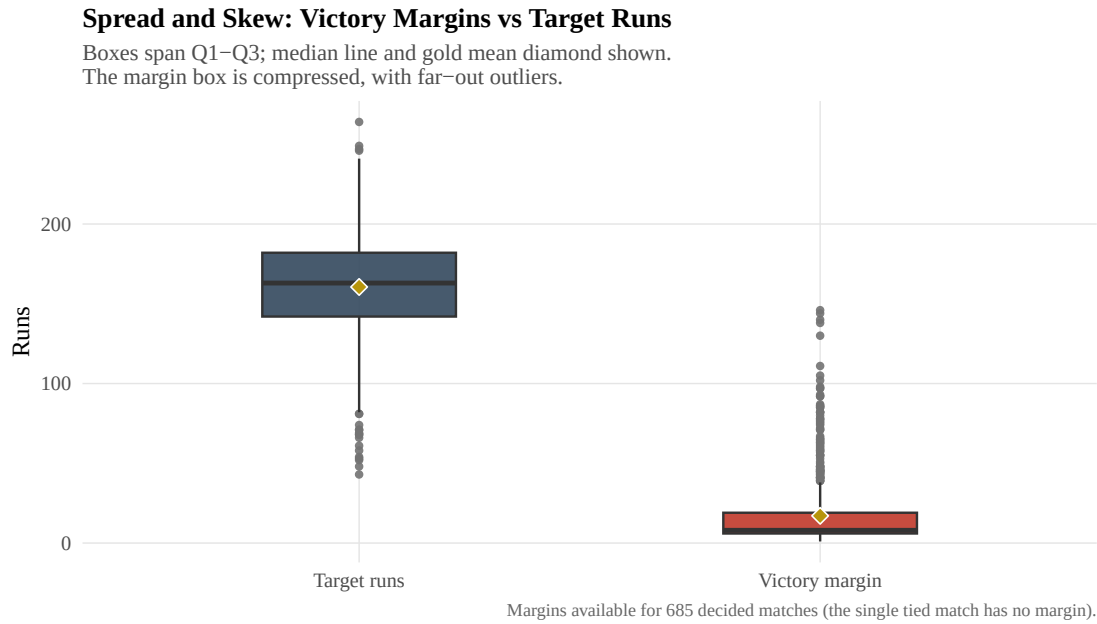


Figure 2: Box-and-whisker comparison of victory margins and target runs. The compressed margin box with its whisker to 146 illustrates the strong right skew described in Section 2.2.

3 Probability Modelling

3.1 The Binomial Model for First-Innings Advantage

A natural *Bernoulli* trial is the outcome of a single match, and the count of wins by a side following a given choice of strategy follows a Binomial distribution. Let p be the probability that a team which chooses to *bat first* wins the match. Across the sample, of the 279 matches in which the toss winner chose to bat first, the same side won 128, giving

$$\hat{p}_{\text{bat}} = \frac{128}{279} = 0.459.$$

In contrast, of the 407 matches in which the toss winner chose to field, the chasing side won 226, so that

$$\hat{p}_{\text{chase}} = 0.555.$$

Assuming a fair coin, $p_0 = 0.5$ for either strategy. The exact two-sided Binomial test of $H_0 : p = 0.5$ against $H_1 : p \neq 0.5$ yields a p -value of 0.188 for batting first — not significant at $\alpha = 0.05$ — but 0.029 for chasing, which *is* significant. The 95% confidence intervals are

$$\text{bat first: } [0.399, 0.519] \quad \text{chasing: } [0.506, 0.604].$$

The chasing interval lies entirely above 0.5, supporting a genuine but modest first-innings advantage for the fielding side in this era.

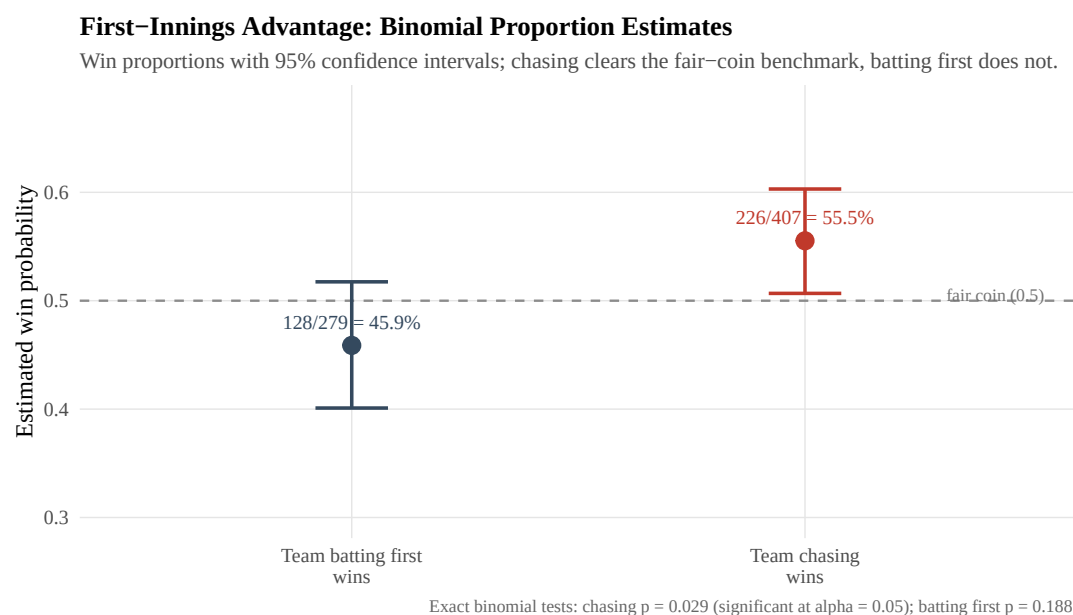


Figure 3: Estimated win probabilities under the Binomial model with 95% Jeffreys confidence intervals. The chasing estimate clears the fair-coin line; batting first does not.

3.2 A Probability Mass Function for Wicket Margins

When a match is won by chasing ($n = 370$), the margin is recorded as the number of wickets in hand, a discrete variable X taking values from 1 to 10. Its empirical probability mass function is shown in Figure 3.2; the most likely outcome is $X = 6$ with $P(X = 6) = 0.205$. The distribution is unimodal and concentrated between 3 and 8 wickets.

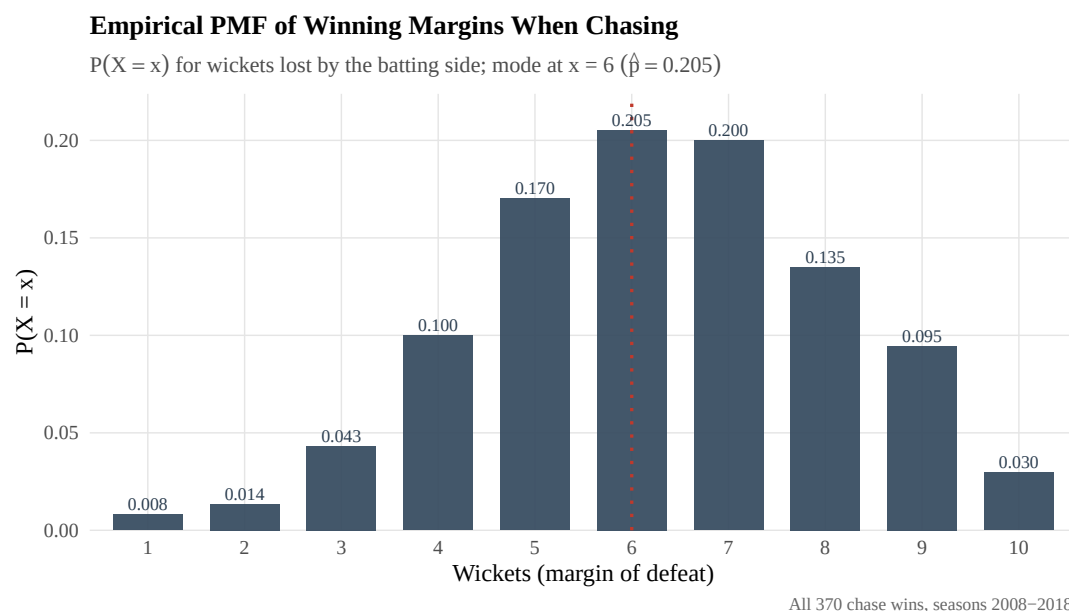


Figure 4: Empirical probability mass function of the wicket margin in the 370 chases won by the fielding side.

3.3 Range-Based Probabilities

Where closed-form integrals of a density are unavailable for empirical data, the probability of an interval is estimated directly from the relative frequency of observations in that interval. For target runs T :

$$\begin{aligned} P(140 \leq T \leq 160) &= 0.233, & P(161 \leq T \leq 190) &= 0.389, \\ P(T > 200) &= 0.087, & P(T < 120) &= 0.093. \end{aligned}$$

Nearly 23.3% of all targets fall in the 140–160 band and a further 38.9% in the 161–190 band, so a team setting a target above 190 enters territory reached in only about 9% of matches. For victory margins, $P(\text{margin} > 50) = 0.074$ and $P(\text{margin} \geq 100) = 0.012$, again illustrating how rare outright domination is relative to close finishes.

4 Regression Analysis

To test whether first-innings targets drifted upward over the decade we fit a simple linear regression of the mean target on season:

$$\hat{y} = a + b x,$$

where x is the season code (1–11). Least squares gives

$$\hat{y} = 153.83 + 1.15 x,$$

implying an average increase of 1.15 runs per season. The fitted model explains 40.5% of the variance in seasonal mean targets ($R^2 = 0.405$), and the slope is significant at the 5% level ($p = 0.035$).

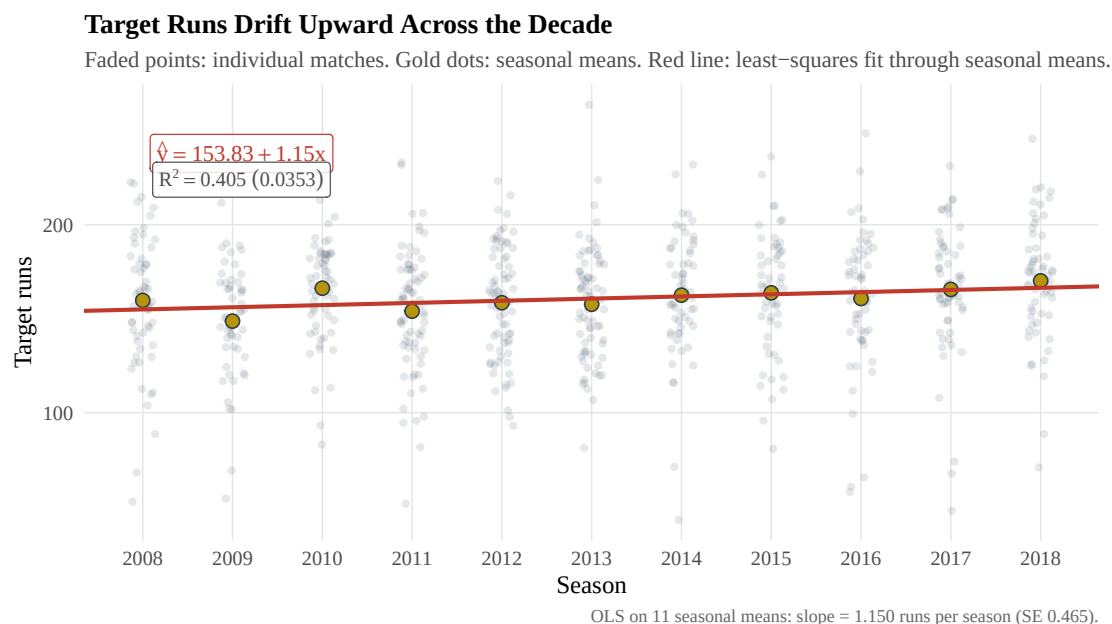


Figure 5: First-innings target vs season. Individual matches are shown faded, the gold points are seasonal means, and the red line is the least-squares fit. The positive slope supports a gradual inflation of scoring across 2008–2018.

5 Conclusion and Reflection

The analysis paints a coherent picture of a decade of IPL cricket. First-innings targets are concentrated near 160–165 runs, are mildly left-skewed, and drift upward by roughly 1.2 runs per season — a statistically detectable increase in scoring. Match margins, by contrast, are highly right-skewed: most games

are decided by a handful of runs or wickets, while blow-outs beyond 50 are rare (about 7% of decided matches).

On the probabilistic side, the Binomial model reveals a genuine first-innings advantage for the chasing side ($\hat{p} = 0.555$, $p = 0.029$), a result consistent with the fielding advantage under floodlights. Team dominance is real but measured: Chennai Super Kings won 90 of their 146 matches ($\hat{p} = 0.616$, $p = 0.006$), a rate the Binomial test confirms against a fair-coin null, while Mumbai Indians ($\hat{p} = 0.571$) fall just short of conventional significance.

The techniques used — descriptive statistics, empirical PMFs, range-based probabilities and the Binomial model — map directly onto the course curriculum and are well suited to match-level data. Their principal limitation is that relative-frequency estimates of probability rest on the assumption that the sample is representative of the broader population of IPL matches; conclusions therefore describe the 2008–2018 era rather than cricket in general. A further limitation is the collapse of many rich covariates (player lists, venues, pitch conditions) into ten variables; a hierarchical model that respected venue and opponent strength would refine the estimates, but lies beyond the scope of this case study.